MATHS MANTHAN
• TUITION CLASSES •
e-mail: contact@mathsmanthan.co.in | Helpline: +91-8171546558

Subject: Mathematics | **WEEKLY CHAPTER TEST – 02 (POLYNOMIALS)** | **Time:** 45 Minutes

Class: Class X (CBSE) | **Chapter 2: Polynomials** | **MM: 20**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Internal choice is provided in Question 10(iii).
2. **Section A** comprises 5 questions (4 MCQs and 1 Assertion-Reason) of 1 mark each.
3. **Section B** comprises 1 VSAQ of 2 marks and 3 SA questions of 3 marks each.
4. **Section C** comprises 1 Applied Case Study with 3 sub-parts of 4 marks total.

SECTION A: Multiple Choice Questions (5 × 1 = 5 Marks)

Q1. Which of the following curves cannot be the graph of a quadratic polynomial? [1]

(A) A parabola opening upward intersecting x-axis at 2 points.
(B) A parabola touching x-axis at exactly 1 point.
(C) A smooth continuous curve intersecting x-axis at 3 distinct points.
(D) A parabola opening downward without intersecting x-axis.

Q2. If one zero of the quadratic polynomial $x^2 + 3x + k$ is 2, then the value of k is: [1]

(A) 10 (B) -10 (C) -7 (D) -2

Q3. A quadratic polynomial, the sum of whose zeroes is 0 and one zero is 3, is: [1]

(A) $x^2 - 9$ (B) $x^2 + 9$ (C) $x^2 + 3$ (D) $x^2 - 3$

Q4. A quadratic polynomial, the sum of whose zeroes is -5 and their product is 6, is: [1]

(A) $x^2 + 5x + 6$ (B) $x^2 - 5x + 6$ (C) $x^2 - 5x - 6$ (D) $-x^2 + 5x + 6$

Q5. **Assertion (A):** The zeroes of the quadratic polynomial $x^2 + 99x + 127$ are both negative.
Reason (R): If all coefficients a, b, c in $ax^2 + bx + c$ have the same positive sign, then all real zeroes must be negative. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false. (D) (A) is false but (R) is true.

SECTION B: Short Answer Type Questions (1 × 2 + 3 × 3 = 11 Marks)

Q6. Find the zeroes of the quadratic polynomial $p(t) = t^2 - 15$ and verify the relationship between zeroes and coefficients. [2]

Q7. If α and β are the zeroes of the polynomial $p(y) = 5y^2 - 7y + 1$, find the value of: [3]

$$\frac{1}{\alpha} + \frac{1}{\beta}$$

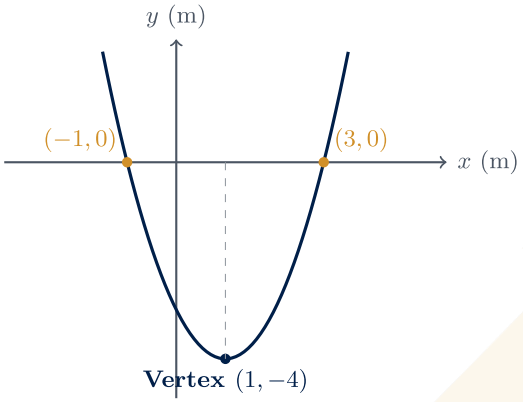
Q8. If $x = 1$ is a zero of the polynomial $p(x) = ax^2 - 3(a-1)x - 1$, then determine the value of a. [3]

Q9. For what value of k is -4 a zero of the quadratic polynomial $x^2 - x - (2k + 2)$? [3]

SECTION C: Applied Case Study Analysis (1 × 4 = 4 Marks)

Question 10: Case Study Analysis — Storm Damage Bent Wire

During a severe tropical cyclone, a roadside metal electrical cable got bent into a smooth parabolic shape. An engineer plots its profile on a coordinate grid where 1 unit = 1 metre. The wire intersects the ground (x-axis) at $x = -1$ and $x = 3$, and its lowest point on the grid is $(1, -4)$.



Based on the above mathematical situation, answer the following questions:

(i) Name the mathematical curve formed by the bent wire and state the number of zeroes of the representing polynomial. [1]

(ii) Write the zeroes of the polynomial represented by the shape of the wire. [1]

(iii) **(a)** Determine the quadratic polynomial expression $f(x)$ representing the shape of the wire. [2]

— OR —

(b) Find the value of the polynomial at $x = 4$ and at $x = -2$. What do these values indicate about the axis of symmetry of the wire? [2]

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